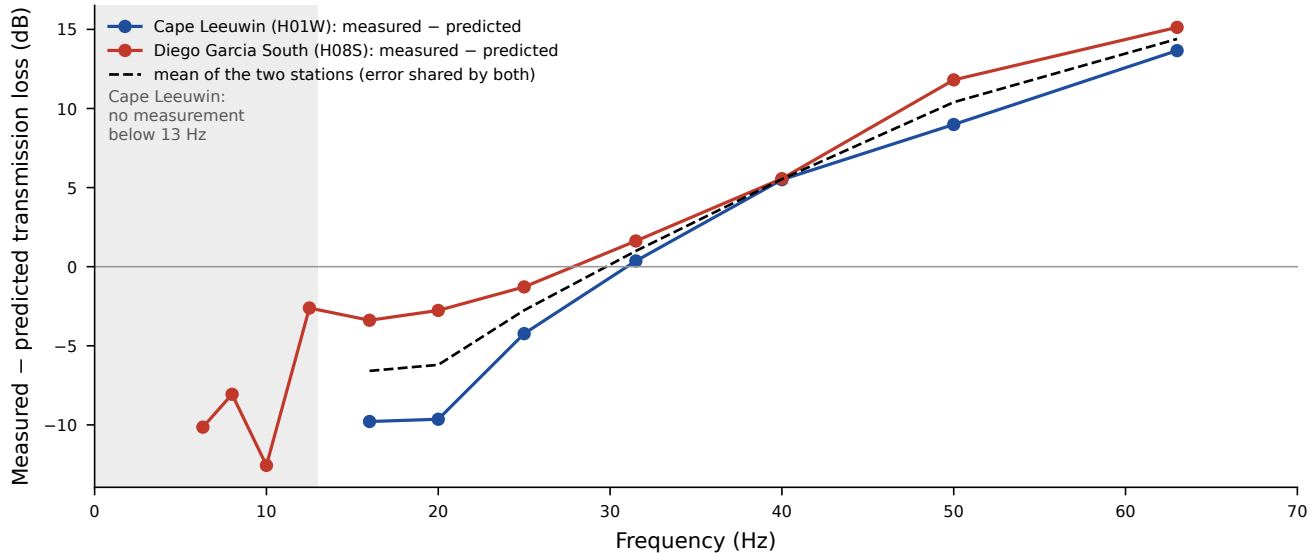


The model's error for the air9 shots tilts with frequency, in the same direction at both stations



What the chart shows: the gap between the sound loss Blackman et al. measured for the 2001 air9 airgun shots and the loss our model predicts, at each frequency. A flat line at zero would be a match. Instead the measured loss is 1–13 dB lower than predicted below 25 Hz and 9–15 dB higher at 50–63 Hz, at both stations. A tilt shared by paths of very different length points to the assumed airgun source spectrum, or to something near the source, rather than to the long-range path. The loss is not validated in its frequency slope.

Technical: residual = Blackman Fig. 23 air9 (digitised, ± 2 dB) – KRAKEN (hard bottom, $z \leq 10$ m) at the model's third-octave centres inside the measured span (H01W 13–70 Hz; H08S 5–70 Hz). Least-squares slope per octave: H01W +12.7 dB, H08S +7.4 dB. The pre-registered rule (median and RMS over 13–40 Hz) gave 'partly validated' and cannot detect a slope; a slope criterion is added for future validation (disclosed, not retroactive). Station difference H01W – H08S: median –1.5 dB, RMS 3.4 dB (hydroacoustics-blackman-validation.md).